

Weil Positivity and the Trace Formula

Formalization Status of the Archimedean Place

arXiv:2006.13771 (Connes–Consani)

Lean 4 / Mathlib formalization project

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Abstract

This document records the current state of the Lean 4 formalization of the archimedean contribution to Weil positivity, following the paper of Alain Connes and Caterina Consani, *Weil positivity and Trace formula – the archimedean place* (arXiv:2006.13771). The project builds cleanly; all proved statements depend only on the standard axioms (`propxt`, `Classical.choice`, `Quot.sound`).

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1 Executive summary

The elementary analytic core of Sections 2–4 of the paper is essentially complete. In particular:

- the convolution $*$ -algebra structure on test functions,
- the explicit form of the archimedean Weil distribution $W_{\mathbb{R}}$,
- the jump of δ' by 2 at $\rho = 1$,
- the essential negativity of $D \circ Q$ on small support,
- the division property that implements the vanishing conditions at $\pm i/2$,
- the Fourier-side formulation of the positivity of $L = D + W_{\infty}$,
- and the inequality $2\theta'(t) + \hat{\delta}(t) \geq 0$ for all $|t| \geq 60$

are theorems.

What remains open of Corollary 2.3(ii) is exactly the compact interval $|t| \leq 60$. The three remaining `sorry`s are:

- (1) `L_positive` — positivity of L via the operator-trace identity,
- (2) `two_thetaDeriv_add_deltaFourier_nonneg` — the inequality on $|t| \leq 60$,
- (3) `LPositivity_of_fourierSide_nonneg` — needs (2) together with a Bochner-type integrability statement.

The highest-value next target is a rigorous lower bound

$$\hat{\delta}(0) \geq \log \pi - \psi\left(\frac{1}{4}\right) \approx 5.3721834.$$

Together with the already-proved reduction that uses monotonicity, this would collapse the remaining compact verification to a single numerical inequality (or to a far smaller interval).

2 List of Lean source files

The project currently contains **36** Lean files under `RequestProject/`:

File	Primary content
Basic.lean	Multiplicative group $\mathbb{R}_+^*$, Haar measure
SineIntegral.lean	Sine integral Si and siDiv
SiPositivity.lean	Positivity of Si on $(0, \infty)$
SiSmooth.lean	Smoothness of the normalized sine integral
SiAsymptotic.lean	Asymptotics of Si
TraceRemainder.lean	Definition and basic properties of δ
DeltaSmooth.lean	Smooth pieces of δ , jump of the derivative
JumpFormula.lean	Integration by parts against a corner function
WeilDistribution.lean	Explicit rational form of $W_{\mathbb{R}}$
Mellin.lean	Convolution $*$ -algebra, Fourier–Mellin transform
Scaling.lean	Scaling representation (stubs)
RemainderBound.lean	Quantitative & strict essential negativity
Positivity.lean	Functional L , division property, small-support corollaries
Parseval.lean	Parseval identity (52), D -half unconditional
FourierSide.lean	θ' , $\hat{\delta}$, Fourier-side inequality
Digamma.lean	Digamma function auxiliaries
DigammaAsymptotic.lean	Vertical-line asymptotics of $\operatorname{Re} \psi$
ThetaGrowth.lean	Effective lower bounds for $2\theta'$, tail positivity
DeltaDecay.lean	Decaying bound $ \hat{\delta}(t) \leq C/ t $
DeltaDecaySharp.lean	Sharper decay estimates
DeltaFourierZero.lean	Analysis of $\hat{\delta}(0)$
DeltaVariation.lean	Variation / modulus of continuity of δ
NearOrigin.lean	Local analysis near $t = 0$
CompactInterval.lean	Infrastructure for the compact case $ t \leq 60$
TaylorBounds.lean	Taylor expansions used in bounds
KernelBound.lean	Bounds on the kernel $Q\delta$
ArchimedeanKernel.lean	Archimedean kernel estimates
ArchimedeanExplicit.lean	Explicit formulae for archimedean contributions
GammaBound.lean	Bounds involving Gamma / digamma
BandEnergy.lean	Band-limited energy estimates
TailEnergy.lean	Tail energy estimates
Effective.lean	Effective numerical constants
FourierSideAnalysis.lean	Further analysis of the Fourier-side density
Sonin.lean	Sonin space (stubs)
Main.lean	Top-level imports
Skeleton.lean	Status document and informal skeleton

3 Key proved results (chronological highlights)

3.1 Convolution algebra and distributions

- Commutativity, associativity and involution compatibility of $\operatorname{convLog}$.
- Explicit formula for $W_{\mathbb{R}}(f)$ (rational weight away from $\rho = 1$).
- Definition of the trace remainder δ , its positivity, symmetry and the jump $\delta'(1^+) = -\delta'(1^-) = 1$.

3.2 Essential negativity

- Identity $D(Q(\xi * \xi^*)) = -2\|\xi\|^2 + \langle \xi * \xi^*, Q\delta \rangle$.

- Quantitative bound on the remainder for support $|\log \rho| \leq a$.
- Strict negativity $\operatorname{Re} D(Q(\xi * \xi^*)) \leq -\|\xi\|^2$ for sufficiently small support.

3.3 Division property and abstract implication

- Explicit inverse of Q via the Green potential with fundamental solution $e^{-|t|/2}$.
- `exists.Qlog_eq_of_vanishesAtHalf`: a continuous test function supported in $[-a, a]$ that vanishes at $\pm i/2$ is of the form QG with G of class C^2 and the same support.
- The implication of Proposition 3.5 / formula (57) holds with only the positivity of L as remaining hypothesis.

3.4 Fourier side

- θ' defined via the digamma function; evenness and the classical asymptotic proved from first principles.
- Decaying bound $|\hat{\delta}(t)| \leq 104/|t|$ for $|t| \geq 60$.
- **Theorem:** $2\theta'(t) + \hat{\delta}(t) \geq 0$ for all $|t| \geq 60$.
- Reduction of the full inequality to monotonicity of $f(t) = 2\theta'(t) + \hat{\delta}(t)$ plus the single bound $\hat{\delta}(0) \geq \log \pi - \psi(1/4)$.

3.5 Parseval

- Unconditional proof of the D -half of the Parseval identity (52).
- The W_∞ -half is carried as an explicit hypothesis `WinftyParseval` (classical archimedean explicit formula).

4 Remaining analytic gaps

Gap	Nature
<code>L.positive</code>	Requires the operator-trace identity $L(f) = \operatorname{Tr}(\vartheta(f)P\widehat{P}P)$ and the continuous integrated form of the scaling representation.
Compact interval $ t \leq 60$	The only remaining obligation of Corollary 2.3 (ii). Highest-value target: rigorous lower bound on $\hat{\delta}(0)$.
Bochner-type integrability	Needed to pass from the Fourier-side inequality to positivity of L for general positive-definite test functions.

The deeper geometric ingredients of the paper (Sonin projection, prolate spheroidal wave functions, compactness of the remainder kernel, Toeplitz discretization of Section 6) remain outside the present scope; they require substantial additional infrastructure (trace-class operators, spectral theory of the pair of projections).

5 Strategic remarks

In the small-support regime actually used by Connes–Consani the contribution of the finite places vanishes identically. Consequently the entire analytic difficulty of the classical Weil positivity criterion is concentrated at the archimedean place. Completing the verification of $2\theta' + \hat{\delta} \geq 0$ on the remaining compact interval is therefore the highest-leverage step toward an unconditional formalization of the archimedean contribution.

The same local analytic engine (with the digamma terms replaced by the appropriate product of Gamma factors) will later serve for Artin L -functions; small-conductor Artin representations can exhibit a negative archimedean density at the origin and therefore require the same compensation mechanism that is being formalized here for the Riemann zeta function.

6 References

- A. Connes, C. Consani, *Weil positivity and Trace formula – the archimedean place*, arXiv:2006.13771.
- A. Weil, *L’avenir des mathématiques*, in *Œuvres Scientifiques / Collected Papers*, Vol. I.
- A. Weil, *Sur les “formules explicites” de la théorie des nombres premiers*, 1952.

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